Subject: Mathematics | **CBSE CLASS 10 – WEEKLY TEST 04** | **Time:** 45 Minutes
Class: Class X (CBSE) | **Chapter 13: Statistics** | **MM: 20**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choices are provided in question subparts.
2. **Section A** comprises 4 MCQs and 1 Assertion-Reason question of 1 mark each.
3. **Section B** comprises 1 Very Short Answer (VSA) question of 2 marks.
4. **Section C** comprises 3 Short Answer (SA) questions of 3 marks each.
5. **Section D** comprises 1 Case Study based question of 4 marks with internal subparts.
6. Use of calculators is not permitted. All rough work and calculation steps must be clearly shown.

SECTION A: Multiple Choice & Assertion-Reason (5 × 1 = 5 Marks)

Q1. In the formula Mode $= l + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2}\right) \times h$, the symbol f_0 represents: [1]

(A) frequency of the modal class
(B) frequency of the class preceding the modal class
(C) frequency of the class succeeding the modal class
(D) cumulative frequency of the modal class

Q2. The construction of a cumulative frequency table is useful in determining the: [1]

(A) Mean (B) Median (C) Mode (D) All of these

Q3. If each observation of a statistical data having mean $\bar{x}$ is multiplied by k $(ke0)$ and then increased by c, the new mean is: [1]

(A) $k\bar{x}$ (B) $\bar{x} + c$ (C) $k\bar{x} + c$ (D) $k(\bar{x} + c)$

Q4. For the distribution: [1]

Marks	Below 10	Below 20	Below 30	Below 40	Below 50
Students	3	12	27	57	75

The modal class is:

(A) $10 - 20$ (B) $20 - 30$ (C) $30 - 40$ (D) $40 - 50$

Q5. **Assertion (A):** The modal class of a grouped frequency distribution is the class having the highest frequency.
Reason (R): Mode is the value of the observation having maximum frequency in a data set. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

SECTION B: Very Short Answer Type (1 × 2 = 2 Marks)

Q6. Find the mode of the following set of observations:
$14, 25, 14, 28, 18, 17, 18, 14, 23, 22, 14, 18$. [2]

SECTION C: Short Answer Type (3 × 3 = 9 Marks)

Q7. Find the missing frequency f if the mode of the following distribution is 65: [3]

Class	20–40	40–60	60–80	80–100	100–120
hline Frequency	10	f	25	16	10
hline					

Q8. Calculate the median marks for 50 students from the following data: [3]

Marks	0–10	10–20	20–30	30–40	40–50
Students	6	11	17	12	4

Q9. Find the arithmetic mean of the distribution using an appropriate method: [3]

Class	100–120	120–140	140–160	160–180	180–200
Frequency	12	14	8	6	10

SECTION D: Case Study Based Question (1 × 4 = 4 Marks)

Q10. CASE STUDY: VOCATIONAL TRAINING INSTITUTE YOUTH SKILL DEMOGRAPHICS [4 Marks]

A state vocational technical council analyzed the age profiles of students and young adults enrolled in digital manufacturing and industrial robotics certification courses.

Age Group (years)	15–20	20–25	25–30	30–35	35–40	40–45
Enrolled Candidates	60	110	80	50	40	20

(i) What is the total number of candidates enrolled and which class has highest density? [1]

(ii) Calculate the modal age of the candidates undergoing training. [2]

(iii) Find the median class interval for the dataset. [1]

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